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Semester - WINTER SEMESTER

Year : 2025 - 26

Slot: A21 + A22 + A23

Course Code – CSA2001

Subject – Fundamentals in AI and ML

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Project Report

Fake News Detection System

1. Introduction

In the modern digital world, information is disseminated quickly via digital platforms. Yet, not all information is correct, resulting in the prevalence of the problem of fake news. The goal of this project is to create a machine learning-based fake news detection tool that categorizes the news as Real or Fake.

It uses Natural Language Processing techniques along with machine learning algorithms like Logistic Regression and Naive Bayes to analyze the data.

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2. Objectives

• To preprocess and clean textual news data

• To convert text into numerical features using TF-IDF

• To train machine learning models for classification

• To compare model performance

• To build an interactive system for user input prediction

• To save and reload trained models for future use

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3. Dataset

- The dataset is stored in a file named news.csv
- It contains the following columns:
    - **title**: Headline of the news
    - **text**: Main content of the news

Data Preprocessing

- Missing values were removed
- Title and text were combined into a new column called **content**
- Text cleaning included:
    - Lowercasing

    - Removing URLs
    - Removing HTML tags
    - Removing punctuation
    - Removing numbers

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4. Feature Engineering

4.1 Text Cleaning Function

A custom function was created using:

- Regular expressions
- String processing

4.2 TF-IDF Vectorization

- Converts text into numerical format
- Removes common English stopwords
- Ignores overly frequent words using max_df=0.7

4.3 Additional Feature

- **Text Length** was added to analyze distribution

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5. Data Labeling

Since the dataset was **unlabeled**, dummy labels were generated:

- 0 → Fake
- 1 → Real

Labels were assigned randomly using NumPy.

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6. Data Visualization

- Bar chart for label distribution
- Histogram for text length distribution

These visualizations helped understand:

- Balance of data
- Variation in content size

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7. Model Building

7.1 Train-Test Split

- 80% training data
- 20% testing data

7.2 Models Used

1. Logistic Regression

- A linear model used for classification
- Handles large feature spaces effectively

2. Naive Bayes (MultinomialNB)

- Based on probability theory
- Works well with text data

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8. Model Evaluation

Evaluation Metrics Used

- Accuracy Score
- Confusion Matrix
- Classification Report:
  - Precision
  - Recall
  - F1-score

Model Comparison

- Both models were evaluated
- The model with higher accuracy was selected as the **best model**

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9. Prediction System

A function predict_news() was created to:

1. Clean input text
2. Transform it using TF-IDF
3. Predict using the trained model

Output

- "FAKE ❌"
- "REAL ✅"

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10. User Interaction

- A loop allows users to enter custom news
- Typing exit stops the program

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11. Model Saving and Loading

Saving

- Model saved as model.pkl
- Vectorizer saved as vectorizer.pkl

Loading

- Models reloaded using pickle
- Verified using a test prediction

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12. Results

- Both models successfully classified news
- Logistic Regression or Naive Bayes selected based on accuracy
- System works interactively for real-time predictions

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13. Limitations

- Dataset was **randomly labeled**, so predictions are not realistic
- No real-world accuracy validation
- Limited feature engineering

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14. Future Improvements

- Use a real labeled dataset
- Apply advanced NLP techniques (like Word2Vec, BERT)
- Deploy as a web application
- Improve accuracy with deep learning models
- Add more features like sentiment analysis

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15. Conclusion

This project shows the application of machine learning and NLP in the detection of fake news. Although dummy labels were used, the system correctly implements the entire process of:

- Data preprocessing
- Feature extraction
- Model training
- Evaluation
- Deployment (basic level)

This project can be a good starting point for the development of a real fake news detection system.

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16. References

- Scikit-learn Documentation
- Python Libraries: Pandas, NumPy, Matplotlib
- NLP Concepts (TF-IDF, Text Cleaning)

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